SIVASAKTHI V

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SUMMARY

Backend and AI systems developer experienced in building scalable FastAPI event pipelines, RAG systems, and integrating LLMs with SMT-based formal verification for deterministic agent safety. Skilled in Apache Spark, Airflow, database concurrency analysis, and cloud architecture, backed by strong algorithmic problem-solving skills.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering, *Chennai Institute of Technology* 2029
CGPA: 8.5 / 10 Current Semester: 3rd Chennai

TECHNICAL SKILLS

Languages
Python, Java, C, C++, SQL, TypeScript, Bash, Javascript

Big Data
Apache Spark (Spark Core, Spark SQL, RDDs), Apache Hadoop (HDFS, MapReduce, YARN), Apache Hive (HiveQL, Hive MetaStore), Apache Sqoop

Databases & Storage
MySQL, SQLAlchemy

Workflow & Systems
Apache Airflow (DAGs, Schedulers), Cron Jobs, Linux (Ubuntu, CentOS, CLI: ps, kill, nohup)

AI & Backend
Z3 / Z3Py, Formal Verification, AI Agent Security, FastAPI, Pydantic, RAG, LLMs

Cloud & Tools
AWS, Git, GitHub, Blend

Web & Frameworks
React, Node.js, Express.js, REST APIs, Tailwind CSS

CERTIFICATES

AWS Certified AI Practitioner

Fundamentals of Analytics on AWS – Part 1 & Part 2

Oracle Cloud Infrastructure AI Foundations Associate

AWS Cloud Practitioner Essentials

CISCO Python essentials 1, 2

Databricks big data certification

Mongo db certifications

PROJECTS

AVE (Asynchronous Verification Engine), *Python, FastAPI, Z3Py, Pydantic, SQLAlchemy*
- Architected an execution guardrail to intercept AI agent tool calls and validate arguments prior to external runtime execution.
- Formulated formal verification rules using the Z3 SMT solver across actions like fund transfers and deletions to return deterministic ALLOW/BLOCK decisions.
- Built modular services including AVEGateway, PolicyEngine, and audit logging to record parameter extractions and policy violations.

DataRace, *Python, Database Concurrency Analysis*
- Designed a concurrency analysis tool to isolate database race conditions and anomalous transaction interleavings that pass standard sequential testing.
- Structured multi-transaction execution scheduling analysis to systematically uncover concurrency-related state inconsistencies.
- LIVE - DATARACE

OXYS, *Python, FastAPI*
- Developed an event-streaming backend pipeline tailored for high-throughput data ingestion and low-latency API request lifecycles.
- Structured asynchronous route handlers and lifecycle hooks for real-time stream processing and automated deployment workflows.
- LIVE - STREAMPULSE

ICHNOS, *TypeScript, Web Technologies*
- Engineered a web application dedicated to trace inspection, execution telemetry monitoring, and structured data workflow visualization.
- Built modular frontend components to render execution traces and provide clear visual monitoring of data flows.
- LIVE : TRACELAKE-V

ACHIEVEMENTS

Winner, BuildFest '26

Finalist, Monad Blitz Hackathon

Finalist, TNWISE CBE

Finalist, Build with DataHub: The Agent Hackathon

Competitive Programming
437+ problems solved on LeetCode across core Data Structures and Algorithms.